

# Bone Age Prediction from Hand Radiographs Using Lightweight CNN Regression and Patch-Based Fusion

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**Abstract**—Bone age assessment from pediatric hand radiographs is a fundamental task in clinical practice, traditionally performed using atlas-based methods that are time-consuming and subject to inter-observer variability. In this work, we study bone age estimation as a regression problem under constraints that require training convolutional neural networks from scratch. We compare a lightweight whole-image CNN baseline with a patch-based framework employing feature-level fusion, and we analyze the contribution of gender metadata as side information. Experiments on the RSNA Pediatric Bone Age dataset show that incorporating gender substantially improves prediction accuracy, and that learned patch-based fusion further reduces validation error at the cost of increased inference time. The results highlight a clear accuracy–complexity trade-off and provide insights into model behavior across age ranges.

**Index Terms**—Bone age estimation, hand radiographs, regression, convolutional neural networks, feature fusion, efficiency

## I. INTRODUCTION

Bone age assessment from pediatric hand radiographs is routinely used in pediatrics to estimate skeletal maturity and to support the diagnosis and monitoring of growth disorders. Conventional approaches such as the Greulich–Pyle atlas require expert interpretation and can suffer from inter-observer variability, motivating automated methods.

In this project, we address bone age estimation as a supervised *regression* problem from hand X-ray images, evaluated primarily by mean absolute error (MAE) in months. Unlike many recent approaches that rely on large pretrained backbones and transfer learning, the ML4HD project constraints require training models *from scratch*. This setting makes the accuracy–complexity trade-off central: improvements should be assessed jointly with computational cost (parameter count and inference time).

We compare two learning paradigms under identical experimental conditions (same dataset splits, preprocessing, optimizer, and early-stopping protocol): (i) a whole-image lightweight CNN regressor and (ii) a patch-based framework that extracts a fixed overlapping grid of patches and combines patch representations via fusion. In both paradigms, we also evaluate the contribution of gender metadata as side information.

### Contributions.

- A lightweight whole-image CNN baseline for bone age regression trained entirely from scratch.

- A patch-based regression model using fixed overlapping patches and two fusion strategies (mean vs. learned softmax weighting) with controlled ablations.
- An empirical study of gender fusion and its impact on validation MAE.
- A systematic evaluation reporting performance (MAE in months) together with efficiency metrics (parameter count and inference time per image).

The remainder of the paper is organized as follows: Section II reviews prior work; Section III describes the processing pipeline; Section IV details the dataset and preprocessing; Section V presents the learning framework; Section VI reports experimental results and analyses; and Section VII concludes with limitations and future work.

## II. RELATED WORK

Bone age estimation from pediatric hand radiographs has been extensively studied using deep learning techniques. Early works demonstrated that convolutional neural networks (CNNs) can achieve clinically relevant accuracy when trained on sufficiently large datasets. Larson et al. [1] showed that CNN-based models outperform traditional atlas-based approaches, particularly when auxiliary metadata such as gender is incorporated.

More recently, the RSNA Pediatric Bone Age Machine Learning Challenge [2] provided a standardized benchmark and highlighted the effectiveness of deep architectures combined with transfer learning and ensemble strategies. Top-performing solutions typically relied on large pretrained backbones (e.g., ResNet or Inception variants), extensive data augmentation, and model ensembling, achieving mean absolute errors of only a few months.

While these approaches establish strong upper-bound performance, their reliance on pretrained models and high-capacity architectures makes them less suitable for constrained learning scenarios. In particular, training CNNs from scratch with limited architectural complexity remains underexplored, despite being relevant for educational settings and for applications where computational resources, labeled data, or pretraining corpora are restricted.

Patch-based and region-focused strategies have also been investigated in medical imaging to emphasize anatomically

informative areas and mitigate background noise. Such approaches often rely on explicit localization, segmentation, or anatomical priors. In contrast, we adopt a simple fixed overlapping patch extraction scheme without external annotations, coupled with feature-level fusion, enabling a direct and controlled comparison with whole-image regression under identical training conditions.

In this work, we position our contribution at the intersection of these directions by systematically studying lightweight CNN regressors trained entirely from scratch, the effect of gender metadata, and the benefits and limitations of patch-based fusion, while explicitly reporting accuracy–complexity trade-offs.

### III. PROCESSING PIPELINE

This section provides a high-level overview of the end-to-end workflow, from input radiographs to bone age prediction. The pipeline is designed to enable a controlled comparison between whole-image and patch-based learning strategies under identical preprocessing and training conditions.

#### A. Whole-image regression

In the whole-image setting, the complete hand radiograph is used as input. Each image is resized to a fixed resolution and intensity-normalized using statistics computed on the training set. The normalized image is processed by a lightweight convolutional neural network (CNN) encoder that extracts a global feature representation. This representation is mapped to a single scalar output representing bone age in months via a regression head. When available, gender metadata is concatenated with the image-level feature vector prior to the final regression layers.

#### B. Patch-based regression with fusion

In the patch-based setting, the resized image is decomposed into a fixed grid of overlapping patches. Each patch is processed independently by the same CNN encoder used in the whole-image model, ensuring architectural consistency. This produces a set of patch-level feature embeddings, one per patch.

Patch embeddings are then aggregated by a fusion module to obtain a global representation used for bone age prediction. We consider two fusion strategies: (i) mean fusion, which uniformly averages patch features, and (ii) learned fusion, which assigns data-driven weights to patches before aggregation. Gender information, when used, is incorporated after patch aggregation by concatenation with the fused representation. This design ensures that whole-image and patch-based models differ only in the input decomposition and fusion mechanism, enabling a fair and controlled comparison.

### IV. DATA AND PREPROCESSING

#### A. Dataset

We use the pediatric hand radiograph dataset released for the RSNA Bone Age Machine Learning Challenge. The dataset consists of grayscale hand X-ray images annotated with

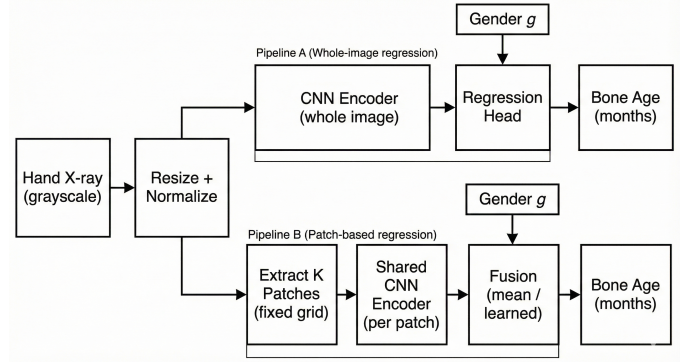


Fig. 1. High-level processing pipeline for bone age prediction. Pipeline A performs whole-image CNN regression with optional gender fusion. Pipeline B extracts fixed overlapping patches, applies a shared CNN encoder, and aggregates patch information through mean or learned fusion, optionally incorporating gender.

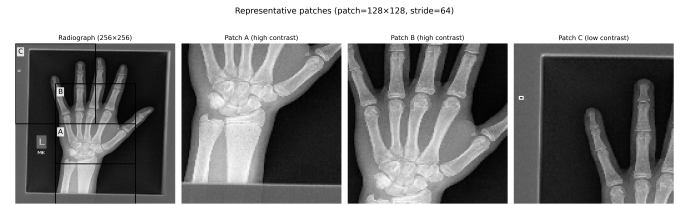


Fig. 2. Example patches extracted from a resized hand radiograph using a fixed overlapping grid (patch size  $128 \times 128$ , stride  $= 64$ ). Some patches capture anatomically informative regions (e.g., wrist and phalanges), while others mainly correspond to background or low-information areas, motivating the use of learned fusion.

skeletal age in months and binary gender metadata. Following the official dataset definition, we use the provided training set (12,611 images) for model learning and the validation set (1,425 images) for performance evaluation. The test set is not used, as ground-truth labels are not publicly available.

#### B. Dataset statistics

Table I reports summary statistics of bone age and gender distribution for the official splits. Figure 3 shows the distribution of samples across age ranges for training and validation. The dataset is imbalanced across ages, with substantially fewer samples in the youngest group (0–24 months). This imbalance is consistent with the higher prediction errors observed for very young patients in Section VI. For visualization, ages above 216 months are assigned to the last bin.

#### C. Preprocessing

All images are resized to a fixed spatial resolution and converted to single-channel intensity values. Pixel intensities are normalized using mean and standard deviation statistics computed on the training set only, in order to avoid information leakage. Target bone age values are standardized using the training-set mean and standard deviation during training and mapped back to months for evaluation. No dataset-specific preprocessing, segmentation, or anatomical annotations are

TABLE I  
DATASET STATISTICS FOR TRAINING AND VALIDATION SPLITS.

| Statistic               | Training      | Validation    |
|-------------------------|---------------|---------------|
| Number of images        | 12,611        | 1,425         |
| Male / Female (count)   | 6,833 / 5,778 | 773 / 652     |
| Male / Female (%)       | 54.18 / 45.82 | 54.25 / 45.75 |
| Mean age (months)       | 127.32        | 127.16        |
| Median age (months)     | 132           | 132           |
| Std. deviation (months) | 41.18         | 41.72         |
| Age range (months)      | 1–228         | 3–228         |

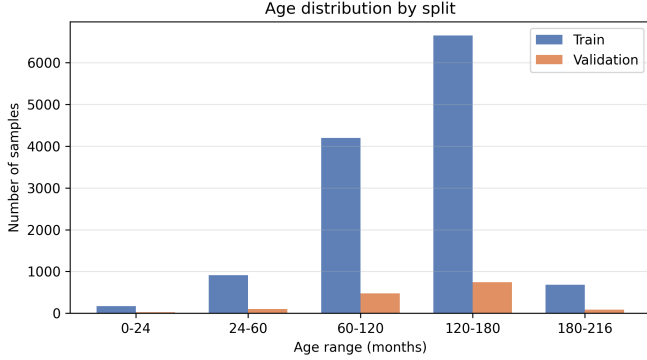


Fig. 3. Distribution of bone age across predefined age ranges for training and validation splits.

used. Unless otherwise stated, preprocessing steps are identical across all models to ensure a fair comparison.

#### D. Training and validation protocol

Models are trained exclusively on the official training split and evaluated on the official validation split provided by the RSNA challenge. Since patient identifiers are not available, no patient-level splitting is performed. Instead, we rely on the predefined split, which preserves similar age and gender distributions across training and validation sets, as reported in the dataset documentation. This protocol matches the evaluation setting of the original RSNA Pediatric Bone Age Challenge [2] and ensures reproducibility.

### V. LEARNING FRAMEWORK

#### A. Problem formulation

Let  $x$  denote a pediatric hand radiograph and  $g \in \{0, 1\}$  a binary gender indicator. The objective is to learn a function  $f_\theta(x, g)$  that predicts the corresponding bone age  $y$  in months. Model parameters  $\theta$  are learned by minimizing a regression loss on the training set. Model performance is evaluated using mean absolute error (MAE) measured in months on the validation set.

#### B. Model architectures

**Whole-image CNN (M1/M2).** In the whole-image setting, the complete radiograph is provided as input to a lightweight

TABLE II  
SUMMARY OF EVALUATED MODEL ARCHITECTURES.

| Model | Input       | Encoder          | Fusion  | Gender |
|-------|-------------|------------------|---------|--------|
| M1    | Whole image | Shared CNN + GAP | –       | No     |
| M2    | Whole image | Shared CNN + GAP | –       | Yes    |
| M3    | Patches     | Shared CNN + GAP | Mean    | No     |
| M4    | Patches     | Shared CNN + GAP | Learned | Yes    |

convolutional neural network. The network consists of stacked convolutional blocks followed by global average pooling, producing a single image-level feature vector. This representation is mapped to a scalar bone-age prediction through a fully connected regression head. In model M2, the gender indicator is concatenated with the image-derived features prior to the final regression layers, enabling demographic conditioning.

**Patch-based CNN with fusion (M3/M4).** In the patch-based setting, each resized image is decomposed into a fixed grid of  $K$  overlapping patches. A shared CNN encoder processes each patch independently, yielding a set of patch-level feature embeddings  $\{\mathbf{z}_k\}_{k=1}^K$ . These embeddings are aggregated by a fusion module to obtain a global representation.

Model M3 employs mean fusion,

$$\mathbf{z} = \frac{1}{K} \sum_{k=1}^K \mathbf{z}_k, \quad (1)$$

which assigns equal importance to all patches. Model M4 uses learned fusion, where scalar importance weights are predicted for each patch and normalized via a softmax operation:

$$\alpha_k = \frac{\exp(w^\top \mathbf{z}_k)}{\sum_{j=1}^K \exp(w^\top \mathbf{z}_j)}, \quad \mathbf{z} = \sum_{k=1}^K \alpha_k \mathbf{z}_k. \quad (2)$$

Gender information, when used, is concatenated with the fused representation before the final regression head.

Table II summarizes the evaluated architectures at a block level. This compact representation highlights the common encoder design and isolates the architectural differences to the input decomposition and fusion strategy, facilitating a controlled comparison.

#### C. Training setup

To stabilize optimization, target bone-age values are standardized using statistics computed on the training set:

$$y' = \frac{y - \mu_{\text{train}}}{\sigma_{\text{train}}}. \quad (3)$$

Models are trained to predict  $y'$ , and predictions are mapped back to months at evaluation time. All architectures are trained using the same optimizer, batch size, learning rate, and early-stopping criterion to ensure a controlled comparison.

#### D. Implementation details

All experiments are implemented in TensorFlow and trained from scratch without pretrained weights. Input images are resized to  $256 \times 256$  pixels. For patch-based models, images are decomposed into overlapping patches of size  $128 \times 128$

TABLE III  
COMPARISON OF EVALUATED MODELS ON THE RSNA VALIDATION SET.

| Model                                 | MAE (months) | Params (M) | Inference (ms/img) |
|---------------------------------------|--------------|------------|--------------------|
| M1: Image only                        | 15.49        | 0.44       | 0.66               |
| M2: Image + gender                    | 11.86        | 0.44       | 0.68               |
| M3: Patches + mean fusion             | 14.21        | 0.44       | 5.38               |
| M4: Patches + learned fusion + gender | <b>10.66</b> | 0.44       | 5.46               |



Fig. 4. Validation MAE comparison across evaluated models (lower is better).

pixels with a stride of 64 pixels, resulting in a fixed number of patches per image.

Models are optimized using the Adam optimizer with a learning rate of  $10^{-3}$  and a batch size of 32, for up to 20 epochs. Early stopping based on validation MAE is employed to prevent overfitting. The Huber loss is used for regression, as it combines the robustness of  $\ell_1$  loss to outliers with the smooth optimization properties of  $\ell_2$  loss, which is desirable given annotation noise in bone-age labels. Mixed-precision training is applied to improve computational efficiency, while model outputs are computed in single precision for numerical stability.

All architectural variants are trained under identical preprocessing and optimization settings, enabling a fair comparison between whole-image and patch-based learning strategies.

## VI. RESULTS

### A. Evaluation metrics

Model performance is evaluated using mean absolute error (MAE) in months on the official RSNA validation split. To quantify the accuracy–complexity trade-off, we additionally report the number of trainable parameters and the average inference time per image (measured with batch inference and repeated forward passes).

### B. Overall comparison

Table III reports the main results. M1 is the whole-image baseline; M2 adds gender metadata; M3 and M4 evaluate the patch-based framework with mean and learned fusion, respectively.

Adding gender metadata yields a large improvement over the image-only baseline (M1→M2), reducing MAE by approximately 3.6 months. The patch-based model with naive mean fusion (M3) performs worse than the whole-image model, indicating that uniformly averaging patch representations is

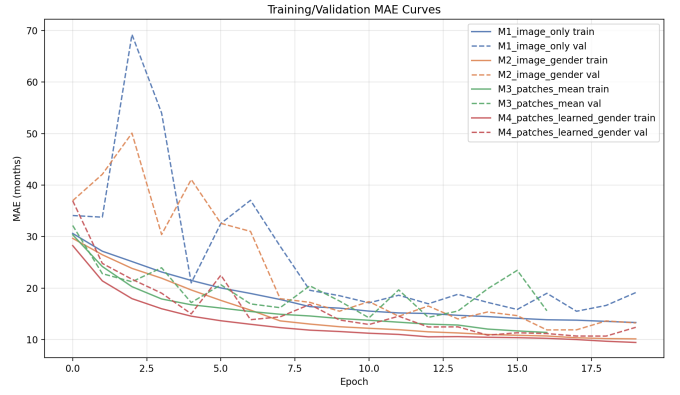


Fig. 5. Training and validation MAE curves for all evaluated models.

not sufficient and likely amplifies background and acquisition artifacts. In contrast, learned fusion with gender (M4) achieves the best validation MAE, suggesting that adaptive weighting of patch information can improve accuracy under the “train-from-scratch” constraint.

Because optimization is stochastic and early stopping selects the best validation epoch, small run-to-run variations (on the order of  $\approx 0.5$ –1 month MAE) were observed across reruns. However, the qualitative conclusions remained stable: gender information improves whole-image regression, and learned patch fusion improves over mean fusion while incurring higher inference cost.

### C. Learning dynamics

Figure 5 shows training and validation MAE curves. Whole-image models converge faster and are computationally cheaper per epoch, while patch-based models are slower due to multiple patch forward passes. Learned fusion (M4) exhibits improved validation behavior compared to mean fusion (M3), consistent with the final MAE results.

### D. Error analysis

To characterize failure modes beyond aggregate MAE, we analyze errors of the strongest whole-image baseline (M2). Figure 6 reports MAE by gender, showing a small but measurable difference between groups. Figure 7 reports MAE across age bins: errors are substantially higher for very young patients (0–24 months), where ossification patterns are less distinctive and the dataset is highly imbalanced, and lower in mid-childhood ranges.

Overall, the experiments highlight a clear accuracy–complexity trade-off: the patch-based learned fusion model (M4) achieves the lowest MAE but requires roughly an order-of-magnitude higher inference time than whole-image models, whereas M2 offers a strong accuracy/efficiency compromise.

### E. Qualitative results

To complement the quantitative evaluation, Figure 8 presents representative qualitative predictions from the validation set for the best-performing model (M4). The figure

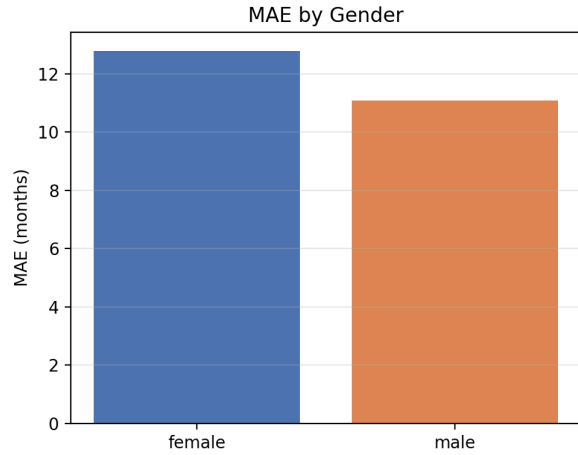


Fig. 6. Validation MAE stratified by gender for model M2.

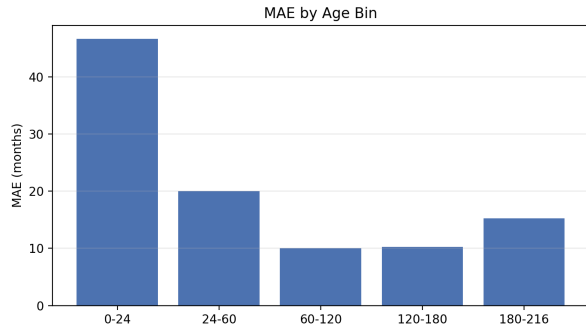


Fig. 7. Validation MAE across age ranges for model M2.

includes examples with low, typical, and high absolute error, together with ground-truth age, predicted age, absolute error, and gender.

Correct predictions generally correspond to cases where ossification patterns of the wrist and phalanges are clearly visible, while larger errors often occur in younger patients or images with less distinctive skeletal structures. These examples qualitatively support the quantitative findings and highlight typical success and failure modes of the model.

## VII. CONCLUDING REMARKS

This project studied bone age estimation from pediatric hand radiographs under the constraint of training convolutional models entirely from scratch. We compared whole-image regression and patch-based fusion strategies and systematically evaluated the contribution of gender metadata. Results show that incorporating gender information substantially improves predictive accuracy, reducing validation MAE by several months compared to image-only regression. Furthermore, learned patch-based fusion achieved the lowest MAE among the evaluated models, at the expense of increased inference time due to multiple patch evaluations.

Qualitative examples on validation set (M4)

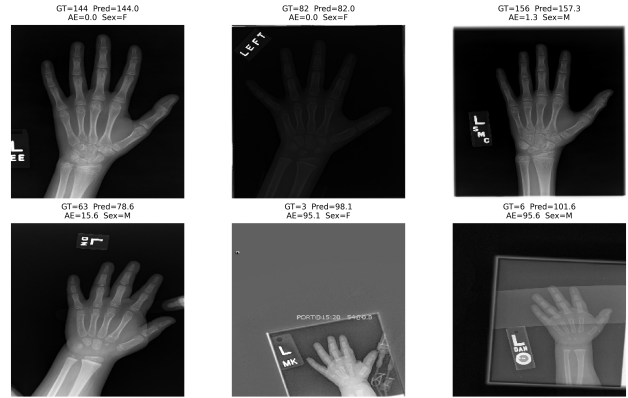


Fig. 8. Qualitative examples from the validation set for model M4, showing representative best, typical, and worst predictions. Each example reports ground-truth age, predicted age, absolute error, and gender.

## A. Limitations

Several limitations should be acknowledged. First, patch extraction is performed using a fixed overlapping grid without explicit anatomical localization, which may introduce sensitivity to image alignment, framing artifacts, and scale variations. Second, prediction errors remain high for very young patients (0–24 months), where skeletal ossification patterns are less visually distinctive and training data are sparse. Finally, the higher computational cost of patch-based models may limit their applicability in latency- or resource-constrained environments.

## B. Lessons learned and future work

From a methodological perspective, this work highlights the importance of controlled ablations and joint accuracy–complexity analysis when comparing learning strategies under constrained settings. While patch-based models can improve accuracy, naïve aggregation is insufficient and learned fusion is required to effectively down-weight uninformative regions. Future work could investigate anatomically informed patch selection, lightweight localization mechanisms, or uncertainty-aware regression to improve robustness. Extensions to semi-supervised or weakly supervised learning may also help leverage additional unlabeled data while preserving the training-from-scratch constraint.

## REFERENCES

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